

v0.21.2 Decision Room Implementation Report

Date: 2026-06-06

This pass upgrades the existing Logits, Softmax, and Sampling Journey cards while keeping Journey order, progress rules, checkpoint randomization, games, dependencies, generated assets, and the one-badge model unchanged.

What Changed

- Bumped the visible app version to `0.21.2`.
- Converted Logits, Softmax, and Sampling to the richer lesson architecture.
- Added lifecycle, durable-vs-temporary, prompt-vs-response, misconception, relationship, and feedback copy.
- Replaced the older Decision Room visuals with coded SVG/HTML diagrams.
- Added dedicated tiny interactions: `logits-raw-toggle`, `softmax-convert`, and `sampling-probability-pick`.
- Updated glossary support for Decision Room terms including probability, decoding step, next token, temperature, top-k, and top-p.

Verification

typecheck passed

build passed

build:pages passed

audit:answers passed

The existing Vite large-chunk warning remains. Screenshot QA at 320px and 390px reported no horizontal overflow.

Card Improvements

Logits

Logits are taught as raw candidate next-token scores from the final hidden state, before probabilities and before choosing. The card now explicitly says logits are not probabilities, permanent memories, or truth scores.

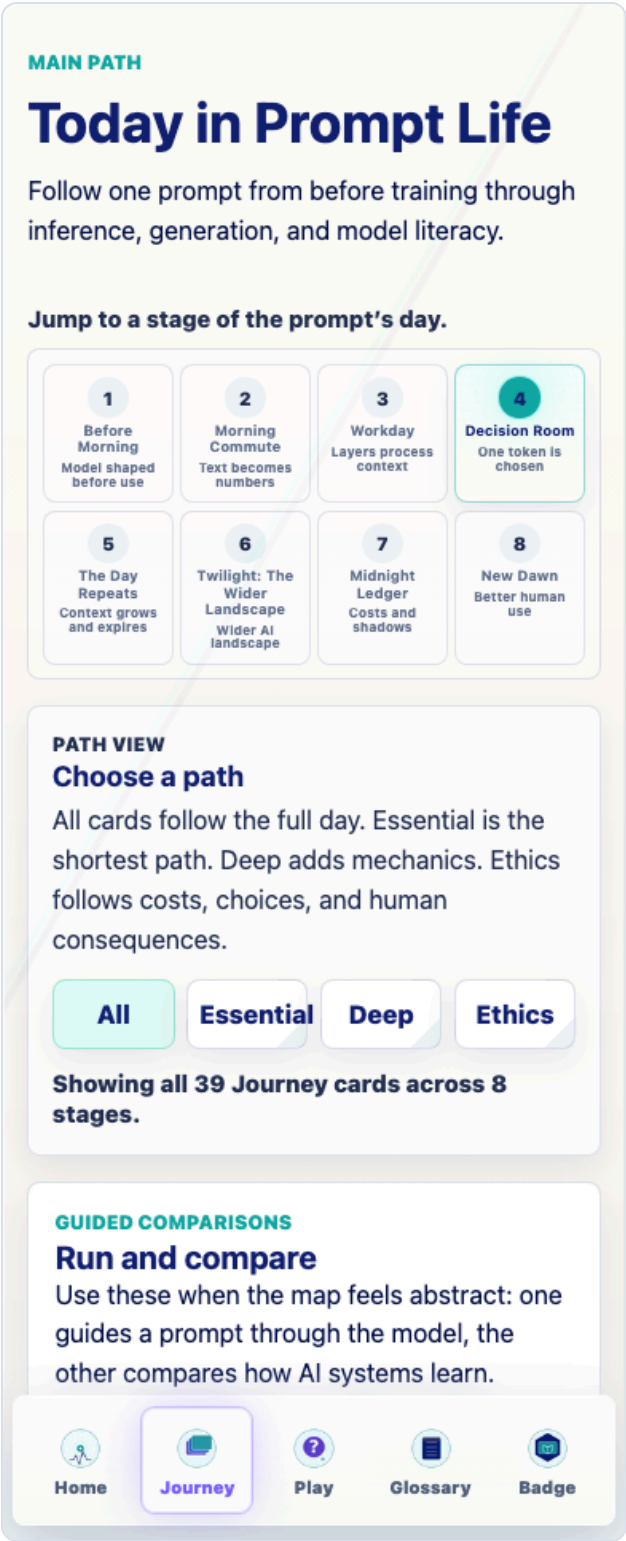
Softmax

Softmax is taught as the conversion from raw scores to probabilities that sum to one. The card repeats that high probability means likely under the model, not guaranteed true.

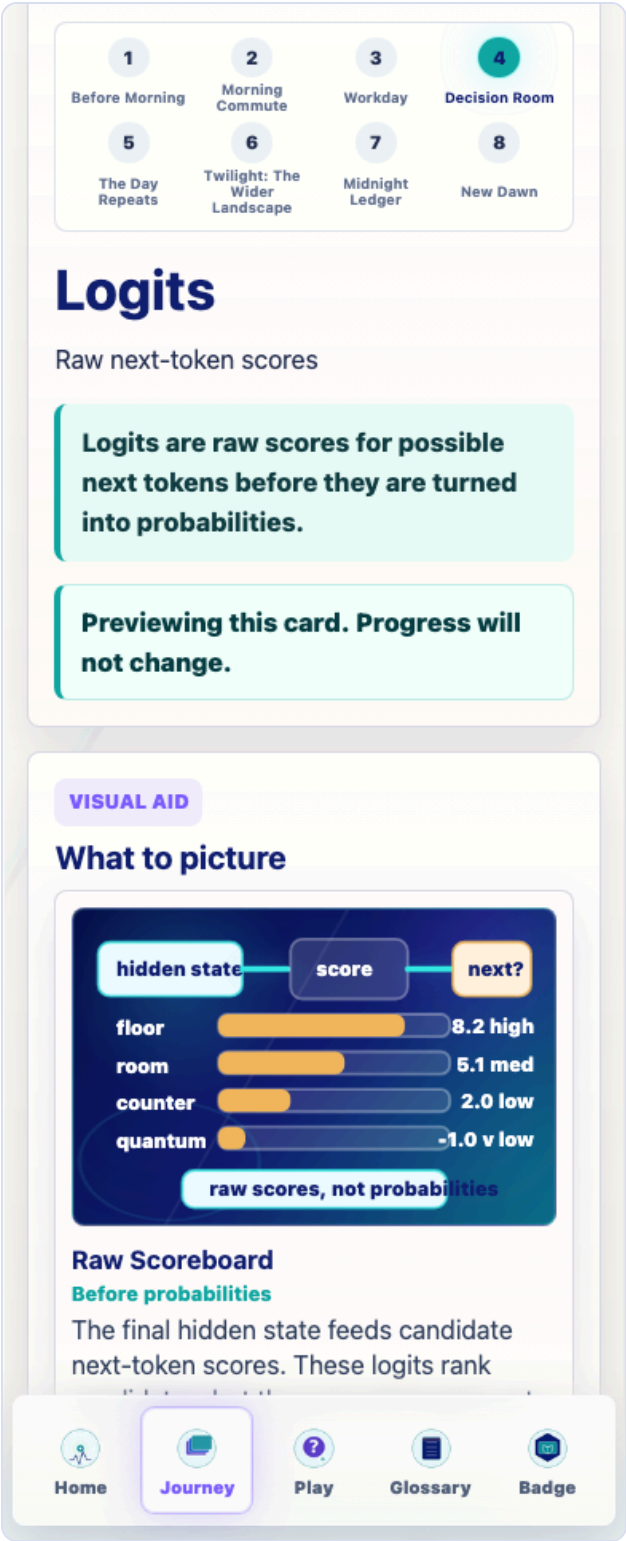
Sampling

Sampling is taught as the decoding step that chooses one generated token from probabilities. The chosen token is appended before the model runs again, and probability is not truth.

Screenshots



Decision Room overview, 390px.



Logits raw score visual, 390px.

TINY INTERACTION

Raw scores or probabilities?

Raw scores

Probabilities?

floor		8.2
room		5.1
counter		2.0
quantum		-1.0

Insight strengthened. Logits are the before state: temporary raw scores for possible next tokens.

Toggle the display and notice that logits are the before state: raw scores, not probabilities or truth.

CHECKPOINT

What are logits?

A

The whole generated answer.

B

Probabilities that sum to one.

C

Raw scores before probabilities.

D

Permanent model memories.

Home

Journey

Play

Glossary

Badge

Logits tiny interaction, 390px.

1

Before Morning

2

Morning Commute

3

Workday

4

Decision Room

5

The Day Repeats

6

Twilight: The Wider Landscape

7

Midnight Ledger

8

New Dawn

Softmax

Scores become probabilities

Softmax is a function that turns raw next-token scores into probabilities that sum to one.

Previewing this card. Progress will not change.

VISUAL AID

What to picture

logits			probabilities	
floor	8.2	softmax		86%
room	5.1			12%
counter	2.0			2%
quantum	-1.0			~0%

probability != truth

sum = 100%

Score to Probability

Normalize before choosing

Softmax converts raw logits into probabilities that sum to 100%. High

Home

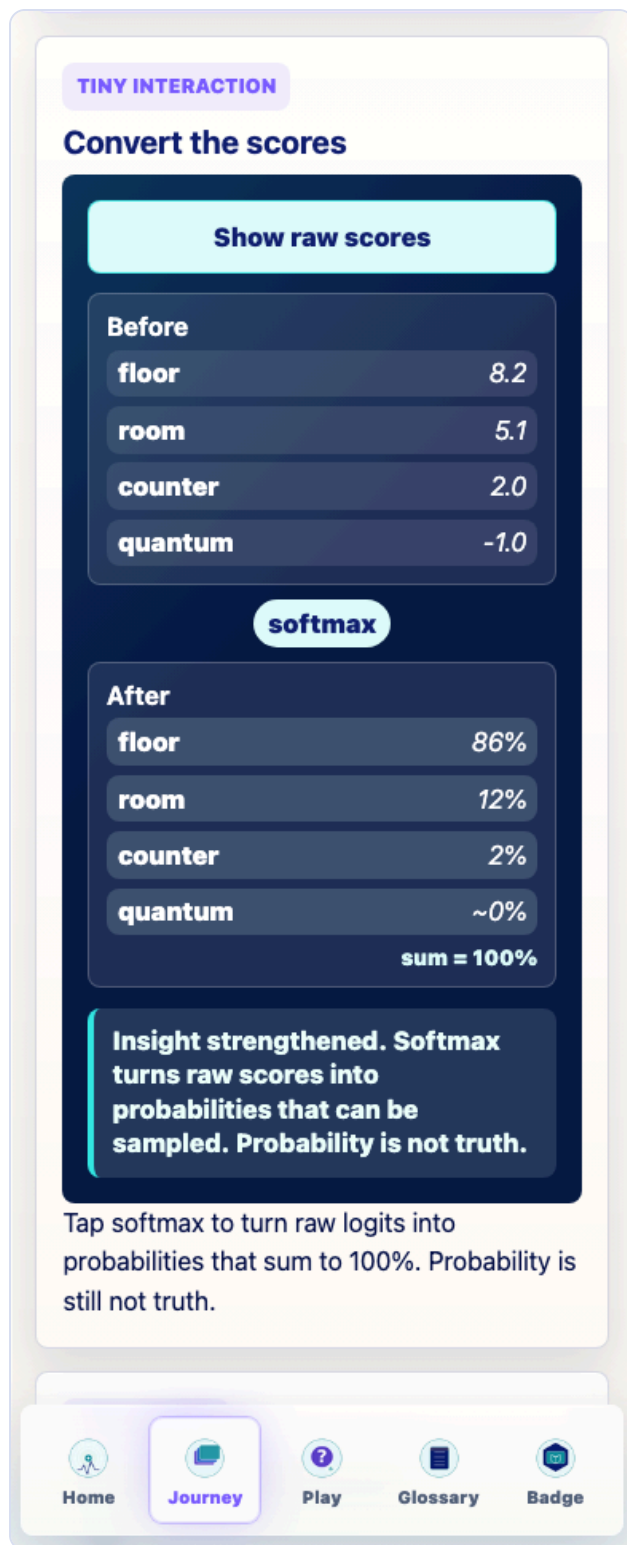
Journey

Play

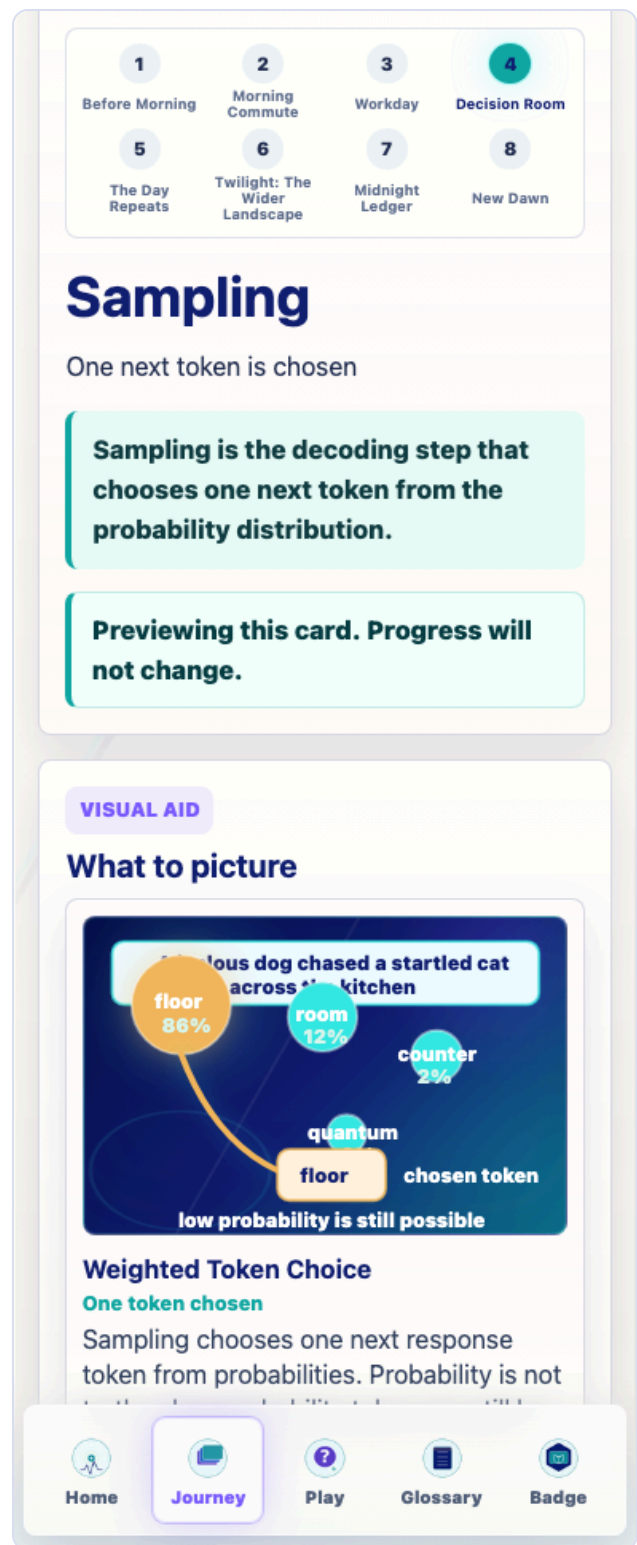
Glossary

Badge

Softmax visual, 390px.



Softmax tiny interaction, 390px.



Sampling visual, 390px.

TINY INTERACTION

Choose from the probability cloud

A jealous dog chased a startled cat across the kitchen...

floor
86%

room
12%

counter
2%

quantum
~0%

Focused: sharper probabilities

More varied: wider possibilities

Which token is most likely in this local context?

Pick the most likely next token from the local context. Probability is not truth; a likely token can still be unsupported in other situations.

CHECKPOINT

When the model chooses one token, what happens next?

A The token is appended to the response-so-far.

B The whole response appears at once.



Home



Journey



Play



Glossary



Badge

Sampling tiny interaction, 390px.

- 1 Before Morning
- 2 Morning Commute
- 3 Workday
- 4 Decision Room
- 5 The Day Repeats
- 6 Twilight: The Wider Landscape
- 7 Midnight Ledger
- 8 New Dawn

Softmax

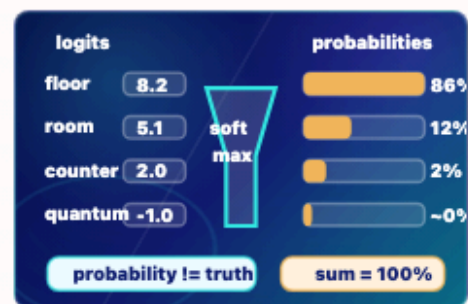
Scores become probabilities

Softmax is a function that turns raw next-token scores into probabilities that sum to one.

Previewing this card.
Progress will not change.

VISUAL AID

What to picture



Score to Probability

Normalize before choosing

Softmax converts raw logits into probabilities that sum to 100%. High probability means likely under the model, not guaranteed true.



Home



Journey



Play



Glossary



Badge

Softmax visual, 320px.

- | | | | |
|----------------------|------------------------------------|----------------------|--------------------|
| 1
Before Morning | 2
Morning Commute | 3
Workday | 4
Decision Room |
| 5
The Day Repeats | 6
Twilight: The Wider Landscape | 7
Midnight Ledger | 8
New Dawn |

Sampling

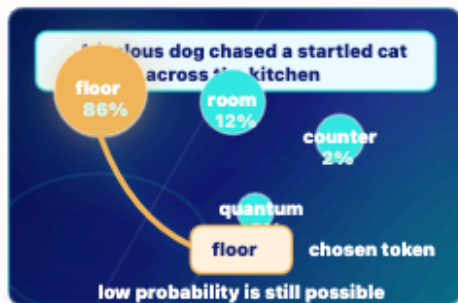
One next token is chosen

Sampling is the decoding step that chooses one next token from the probability distribution.

Previewing this card. Progress will not change.

VISUAL AID

What to picture



Weighted Token Choice

One token chosen

Sampling chooses one next response token from probabilities. Probability is not truth: a low-probability token can still be possible, and a likely

- | | | | | |
|------|---------|------|----------|-------|
| Home | Journey | Play | Glossary | Badge |
|------|---------|------|----------|-------|

Toggle the display and notice that logits are the before state: raw scores, not probabilities or truth.

CHECKPOINT

What are logits?

- A** Probabilities that sum to one.
- B** The whole generated answer.
- C** Permanent model memories.
- D** Raw scores before probabilities.

REFLECT

Teach it back in one sentence.

Reflection is available in Learn mode.

[Return to Journey](#)

- | | | | | |
|------|---------|------|----------|-------|
| Home | Journey | Play | Glossary | Badge |
|------|---------|------|----------|-------|

Logits randomized checkpoint, 390px.

Sampling visual, 320px.

Readiness

No Image 2 asset is recommended for Decision Room. The coded visuals are the right format because the exact labels, scores, probabilities, and truth-boundary language need to remain accessible. Before Morning, Morning Commute, Workday, and Decision Room are ready enough for the next stage audit/implementation pass: The Day Repeats.